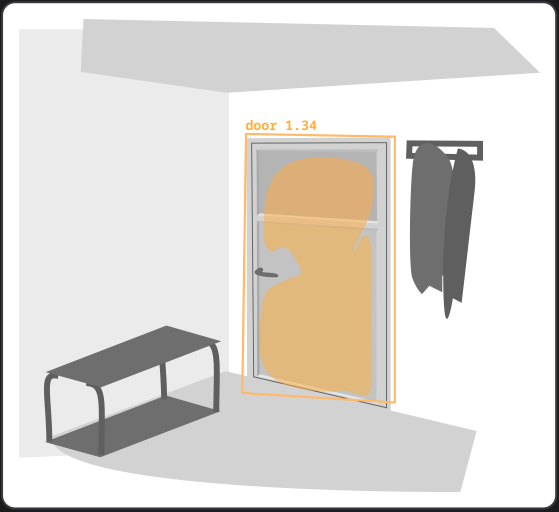
Help for Visually Impaired with Auditory Navigation — Technical Report

26.10.2026
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Project Report for DATA11001
University of Helsinki

0. Introduction



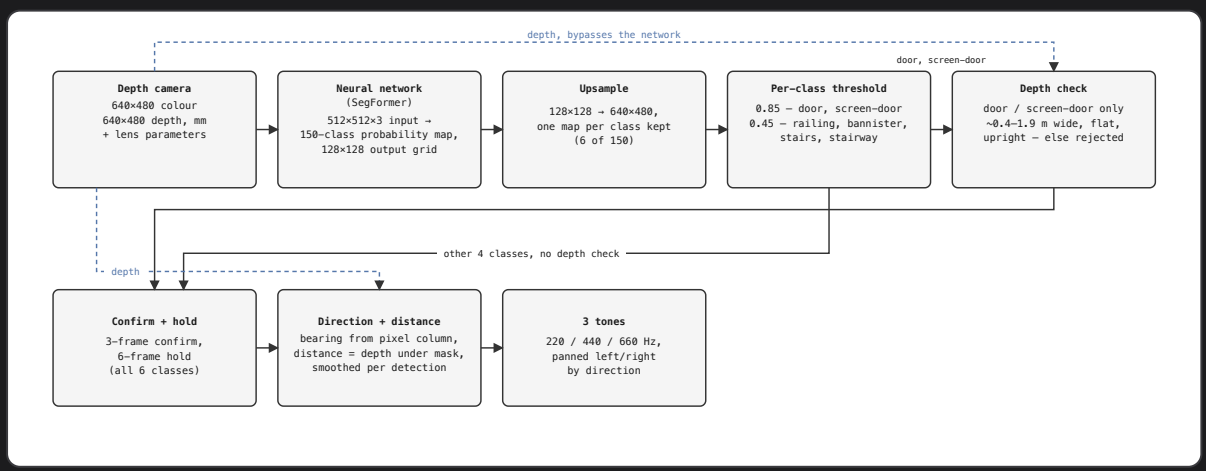
Several assistive devices already help a blind or visually impaired user get around safely. There are smart canes, obstacle-avoidance wearables, sensors that warn about what is in the way, some of them providing a real improvement in everyday ease of life. They share one focus: what to avoid. This project points the same kind of hardware at the opposite question — not what is blocking the path, but what is useful on it. A door, a railing, a flight of stairs: not hazards, but the landmarks a person actually navigates by. This is a proof of concept for finding those, with a depth camera and a neural network, worn or carried by the user.

A screen is no use here, and braille is too slow for something that has to update as the user walks. Natural language's directional vocabulary is coarse. Sound in a form of signal tone is the only channel that fits. It reaches the user continuously, without taking their hands or attention away from walking, and it already carries direction and distance naturally. A person does not have to learn what "to the left" actually means.

This is a proof of concept, not a finished product. A depth camera watches the environment the user is walking through. The system finds doors, railings, and stairs in what it sees, and turns each one into a sound signal, a tone that comes from the right direction, at a loudness that tells the user how far away it is.

This report follows one frame of data on that trip: camera, network, a few filtering steps, and out as sound. It does not try to cover every file or every flag, the project's `README.md` does that. Each section below ends with what the data looks like by the time it leaves that stage, so its shape is traceable all the way through: raw colour and depth in, one short tone out.

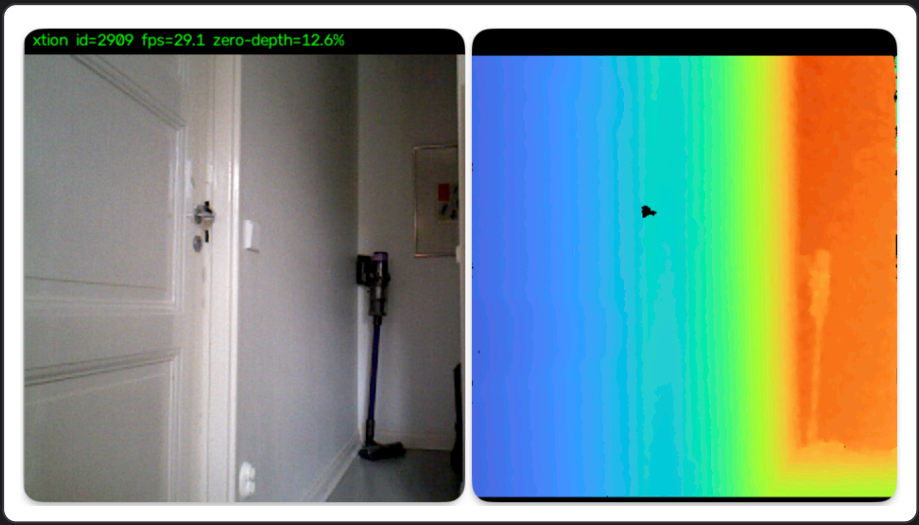
Pipeline at a glance



Two things to notice: the depth check only runs for door and screen-door. The other four classes skip it and reach the tracker directly. And depth never enters the neural network at all (dashed lines). It travels alongside the colour image, straight from the camera to the stages that need it: the depth check, and distance.

1. Where the data starts

One frame starts as two images and a few numbers.



The colour image is 640×480 pixels, 3 channels, one byte per channel (red, green and blue, stored in that order, not OpenCV's usual blue-green-red; the display code flips it just before showing it, not before using it). The depth image is the same 640×480 size but one channel: a 16-bit number per pixel, in millimetres, where 0 means no reading. The two are pixel-aligned: pixel (x, y) in one is the same physical point as pixel (x, y) in the other. That alignment is not automatic: the depth sensor and the colour lens sit a few centimetres apart on the device, so one has to be remapped into the other's point of view before the two line up. That work is already done by the time a frame reaches this pipeline.

A real frame, side by side: colour on the left, depth colourised on the right, near shaded blue and far shaded orange so the gradient reads at a glance. The small black spot partway up the depth image is a zero-depth pixel. It is chrome door handle and the sensor simply could not measure it due to reflectivity. A handful of pixels like that are normal in almost every frame; what matters later is whether they land inside something the system is trying to measure.

Alongside the two images: the camera's lens parameters which are focal length and optical centre, in pixels, one pair per axis (fx, fy, cx, cy). These do not describe the scene; they describe the camera itself, and they're what later turns "pixel column 400, depth 2.3 m" into an actual angle and a real-world sideways offset. There's a frame id, so detections can be matched up frame to frame, and a timestamp.

The depth sensor also has a real, physical range: roughly 0.8–5 m. Closer than that, farther than that, or pointed at glass or a mirror, a pixel just reads zero — no reading, not a wrong one. A door's own glass panel can blank out part of its own detection this way. That gap in the data does not get patched here, since it is a hardware related issue.

This frame comes from one of two sources, and they're indistinguishable once it arrives: a live Xtion camera, or a previously recorded session played back with `segformer replay`. Nothing downstream can tell which, so a recording can stand in for the camera during development.

What leaves this stage: one frame (colour image, depth image, lens parameters, id, timestamp). Nothing has been computed yet.

2. Going over the wire

The frame is packed into a single message: the header fields (id, timestamp, intrinsics, …) plus the colour and depth arrays, each as its raw bytes alongside its dtype and shape, so the other end can rebuild the exact same array without guessing. Nothing is resized, recoloured, or otherwise changed. By default nothing is compressed but an lz4 option exists for it, and it is quite useful since uncompressed feed is 44 megabytes per second.

This goes out over ZeroMQ as a publish/subscribe stream, one message per frame. The consumer side keeps only the newest one: if it is slower than the camera, older frames are dropped instead of getting queued up. Because of this there is no backlog to catch up on.

What leaves this stage: the same frames serialized over network thread.

3. Into the network

Only the colour image goes into the network (depth sits this stage out entirely). It's resized from 640×480 down to 512×512 (bilinear), scaled to the 0–1 range, then normalized with fixed ImageNet statistics (the same mean and standard deviation the model was trained against, one triplet each for red, green, blue). This exact recipe is what HF's own SegFormer preprocessor does by default. Matching it is not optional, since the network was trained against images prepared that way.

Either backend for the SegFormer, MLX or PyTorch, runs the identical recipe and the same network weights. The backend choice only changes which hardware does the arithmetic, not the numbers that come out (see "Known limits" in the project's `README.md` for the small amount of float drift that exists between the two anyway).

The network itself is a 150-class classifier (trained on ADE20K's full label set) run per pixel at its own native, coarse 128×128 grid (the architecture downsamples by 4×, regardless of model size). A softmax turns that into a probability per class per pixel; only the 6 classes this project tracks (door, screen-door, railing, banister, stairs, stairway) are kept, the other 144 are discarded here.

What leaves this stage: a probability, 0 to 1, for 6 classes, at every point of a 128×128 grid. Not yet a detection, and not yet at the frame's own resolution.

4. From probabilities to shapes

Up to here the data is a grid of numbers: a probability per pixel, per class, not yet a shape. Turning that into something usable happens in four steps, run once per class.

First, the 128×128 grid is stretched back to the frame's own 640×480, the same bilinear blur used earlier to fit it to the network. A mask's edge is never sharper than that stretch.

Second, each pixel is kept or dropped by a threshold: 0.85 for door and screen-door, 0.45 for the rest. Argmax ("highest score wins, whatever the number") is not used here, since on a door it eats into the wall beside it.

Third, door and screen-door only. The raw output usually comes back as several disconnected blobs for one physical door. Blobs at the same depth merge into one mask per door, so a second door farther away stays separate. Each merged shape also has to pass a depth check: roughly door-sized (0.4–1.9 m wide), flat, upright, tolerant of a door only partly in frame. Failure of either checks results to shape being dropped. Railing, banister, stairs, and stairway skip this step; nothing depth-based filters them today.

Fourth, a shape has to persist: three frames running to be reported, lingering six frames after it is lost, so one missed frame does not flicker a detection on and off.

What leaves this stage: a list of detections, each with a stable id, class, confidence score, mask, and box, the door and its orange outline in §0's figure, drawn.

5. Turning a detection into a place in space

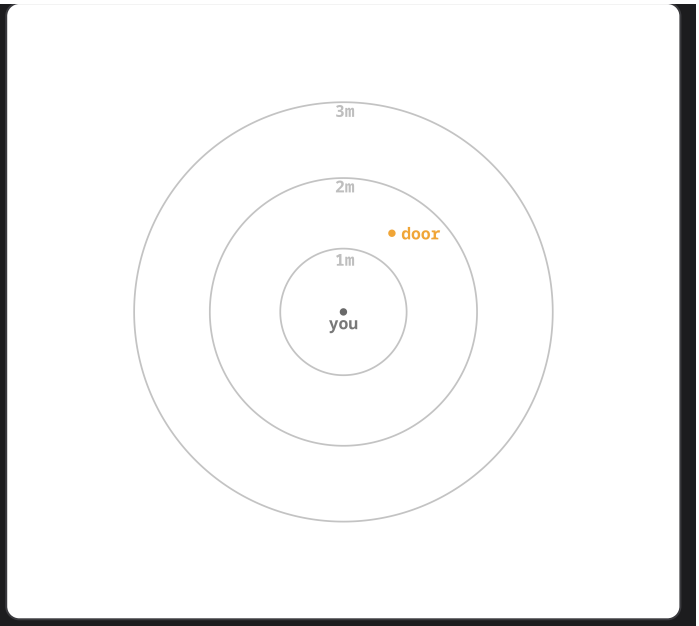
A detection, at this point, is a shape in a 2D image. Converting it into a direction and a distance is the responsibility of this stage, not the network, which never receives depth as input.

Distance is derived from the depth image, sampled within the detection's mask. The mask's outer rim is eroded by five pixels first, since the upsampling in §4 leaves that rim blurred and liable to include background pixels. If the eroded region contains at least fifty valid depth readings, their median becomes the object's distance. Below that threshold, distance is recorded as unknown rather than estimated.

Direction, or bearing, is computed from the horizontal pixel position of the detection and the camera's lens parameters (focal length and optical centre), using the same pinhole relationship introduced in §1. This value does not depend on a successful depth reading: bearing remains available even when distance does not.

Both values are smoothed per detection across frames using an exponential moving average, so a single noisy reading does not cause a visible jump. A detection that stops appearing is retained briefly, then discarded if it is not seen again within ten updates.

What leaves this stage: a cue per detection, a bearing in radians, a distance in metres (or none), and the detection's identity, class, and label, carried forward from §4.



6. Turning a place in space into a sound

This is the final stage, and the one the preceding six exist to serve: a cue, consisting of a class, a bearing, and a distance, becomes something audible.

Each of the three sound classes is assigned a fixed pitch: 220 Hz for door, 440 Hz for railing, 660 Hz for stairs. Where more than one object of a class is in view, the nearest one determines the tone for that class; the others do not add a second tone.

Loudness encodes distance. A detection within one metre is played at full volume; beyond that, loudness falls off with distance. A detection with no usable depth reading, from §5, is still played, at a fixed, reduced volume, so that bearing remains communicated even without a distance.

Direction is encoded twice, not once. Volume differs between the left and right channel according to bearing, and in addition, the near channel leads the far channel by a short interval, on the order of a millisecond. This second cue, an arrival-time difference rather than a volume difference, is closer to how human hearing actually localises a source, and was added because volume alone read as flat for small angles.

The three classes may be muted individually, by pressing 1, 2, or 3. Since the intended user cannot see a screen, this action is confirmed by sound, not by text: a single descending tone for mute, two ascending tones for unmute. A visual indicator exists as well, for a developer at a terminal, but it is not the primary channel.

The keyboard is a placeholder. There is no keyboard on a device the user would actually wear or carry; 1, 2, and 3 stand in for whatever physical input that device will have, such as a small set of buttons. Any such signal can drive the same mute logic with a small code change, since the toggle itself does not depend on how it is triggered.

7. Table 1. Summary of data manipulation stages

Stages of the data manipulation, and the properties of the data that leaves that stage (consequently entering the next stage) are given.

§	stage	data, leaving
1	capture	one `Frame`: 640×480×3 colour, 640×480 depth, intrinsics
2	wire	the same frame, serialized to one message
3	network	probabilities, 6 classes, 128×128 grid

§	stage	data, leaving
4	filtering	a list of detections, each with an id, a mask, a box
5	localisation	a list of cues: bearing, distance (or none), per id
6	audio	a continuous stereo signal

8. Open gaps and reflection

The depth source is the one part of this journey that most plainly needs replacing. The Xtion is a structured-light sensor: it projects an infrared pattern and reads its distortion, a method that fails outdoors in direct sunlight, where ambient infrared overwhelms the projected pattern. A device meant to be worn outside needs a depth source that does not share this failure mode, such as a lidar sensor, which measures distance directly by time of flight and does not depend on a projected pattern at all.

This change is confined to the capture side. §1 established that a frame is just a colour image, a depth image, and a small set of parameters describing the sensor, and §2 that this is serialized into one message and published over ZeroMQ without interpretation. Nothing downstream of that publication, from the subscriber in §2 onward, inspects where the depth values came from. Replacing the Xtion with a lidar sensor means producing the same `Frame` contract from different hardware; it does not require a change to the network, the filtering stage, the localisation stage, or the audio stage.

In terms of design process and planning, we faced issues in determining who this project would serve. We suspected from the beginning that we were unlikely to create an application usable by the target audience, because of challenges relating to developing for consumer products like mobile phones. So we decided to stick with the hardware we had, and focus more on the conceptual benefits this project may provide. We thought that this might be useful for various robotics applications, but decided to limit our focus to navigation for people. This also lead us to make our final delivery a website in which we present our project. We aimed to make the website easily accessible to screen readers, and kept it structurally simple for easy navigation through the text blocks. The website and reading experience could be further improved by adding accommodations for low-vision readers, such as a menu for adjusting the font size and style. This project has been a valuable learning experience for data science, project design and organisation for all of us.

Claude Sonnet 6 was used to polish, improve and language-check this report.